ROOT CAUSE UNLOCKED · CONDITION KIT

The Gut Barrier Root Cause *Kit*

Intestinal permeability is real physiology. The tests sold to measure it vary enormously in how well they have been validated, and knowing which is which is most of what this guide is for.

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The Gut Barrier Root Cause Kit

BEFORE YOU START

Contents

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Seek medical care now for blood in the stool, black tarry stool, unintended weight loss, persistent vomiting, fever with abdominal pain, severe or worsening abdominal pain, or anemia. Those are not leaky gut. They are reasons for a gastroenterologist, and attributing them to intestinal permeability is how a real diagnosis gets missed.

Key 2 applied: what actually breaks the barrier

Unlock Your Gut devotes its second key to what damages the barrier in the first place, and that list is the spine of this whole kit. A repair protocol that never removes the thing doing the damage is a repair protocol you will be repeating.

What breaks it	Why it matters	What to do about it
Medications	NSAIDs, repeated antibiotics, and long-term acid suppression all act directly on the barrier or the community living behind it	Review the list with your prescriber. Some are necessary and some are habit
Dysbiosis	The bacterial community is part of the barrier, not a passenger behind it	Stool analysis where symptoms justify it, then rebuild rather than only kill
Diet	Low fiber diversity, high ultra-processed intake, and alcohol	Covered in the diet guide, which is the longest lever here
Stress	Not a metaphor. Measured directly in humans, see below	Daily down-regulation, treated as part of the protocol
Sleep	Repair happens overnight, and the gut-brain signalling runs both ways	A fixed sleep window, protected like a medication
Infection	An acute gut infection is one of the best-documented ways this starts	History matters. Post-infectious onset changes expectations

Notice how many of those are not supplements. That is the honest shape of gut repair, and it is why the supplement guide is not the first document in this kit.

The honest starting position

Intestinal permeability is real. The barrier exists, it can be measured, it is affected by medications, alcohol, and specific dietary components, and it is altered in several diseases. None of that is fringe.

The market that grew around it is a different matter. Several tests are sold as confirming leaky gut. Their validation ranges from genuinely good to essentially absent, and the ones sold hardest are not always the ones with the most behind them.

This guide sorts them. That is more useful to you than another explanation of tight junctions.

The markers, ranked by what actually backs them

Read your panel as a pattern, not as a verdict. These markers measure different things, and treating them as several confirmations of one fact overstates what you have.

IgA anti-actin, the best validated on the panel

In a multicentre study, IgA anti-actin antibodies showed **80% sensitivity and 85% specificity** for celiac disease with villous atrophy, with an area under the curve of 0.873.[1] Anti-actin levels tracked with the severity of intestinal damage.

The honest caveat: that validation is specifically for celiac disease with villous atrophy, not for permeability in general. In a broader celiac population a separate study found much weaker performance, 41.3% sensitivity and 71.4% specificity, and detection was lower in people already following a gluten-free diet.[2]

So it is a good marker of intestinal damage severity in a specific disease, and a weaker one outside it. That is still the strongest thing on the panel.

LPS and endotoxin antibodies, measuring something different

These reflect bacterial products reaching systemic circulation. In liver disease across all stages, gut-vascular barrier and bacterial translocation markers including endotoxin-core IgA antibodies correlated with each other **but not with intestinal damage markers**.[3] In a cystic fibrosis study, LPS antibody showed reliable intraindividual stability.[4]

Why this matters for reading your report. Translocation and mucosal damage are different processes. An elevated LPS antibody alongside a normal damage marker is not a contradiction to explain away, it is two different findings.

Serum zonulin, where the caution has hardened into a verdict

The assay problem is worse than a failed correlation: the most widely used commercial zonulin ELISA was shown not to detect zonulin at all. An analysis of the kit found it does not detect pre-haptoglobin 2, the molecule zonulin actually is, and instead recognizes other proteins, including properdin, with measured concentrations failing to track the haptoglobin genotypes that determine whether a person can even produce zonulin.[11] Separately, serum zonulin by commercial kit failed to correlate with the physiologic lactulose-mannitol measure of permeability ($r^2 = 0.004$ in 39 first-degree relatives of Crohn's patients), or with intestinal fatty acid binding protein.[5]

The position this kit takes, plainly: skip zonulin testing, serum or stool, until a validated assay exists. The zonulin molecule's biology is real science. The number sold under its name is not a validated measurement of it, and this practice will not interpret one as a diagnosis. If a panel you already ran includes a zonulin value, treat it as uninterpretable rather than as evidence in either direction. Note also that serum zonulin protein and **anti-zonulin antibodies** are different measurements; panels vary in which they run, and the assay critique above concerns the former.

And the equally important other half, because a broken test does not un-discover the physiology. Intestinal hyperpermeability is real, mainstream gastroenterology: a review in the journal Gut details the barrier's components, the validated research measures of permeability (probe-molecule tests like lactulose-mannitol, and tissue methods), and the stressed states in which barrier function measurably deteriorates, while cautioning that public-domain "leaky gut" claims need confirmation before exclusion diets or repair supplements are endorsed.[12] Everyone has intestinal permeability; you could not absorb nutrients without it. The clinical question is excess, exactly as the clinical question in hypertension is not whether you have blood pressure. Anyone who dismisses the entire territory because a bad commercial test exists has confused the assay with the physiology.

Occludin antibodies

I searched for human validation of occludin antibodies as a barrier marker and did not find a comparable body of work. That is not a claim they are useless. It is a statement about what I could and could not verify, and you should weight the result accordingly.

The lactulose-mannitol test, the physiologic standard with a practical problem

This is the closest thing to a direct functional measure: you drink two sugars and their recovery ratio in urine reflects barrier function.

Its problem is variability. In a study running 6,602 tests across 8 countries, mean mannitol recovery ranged from 1.34% to 5.88% between sites, lactulose recovery from 0.19% to 0.58%, with strong sex differences and age effects.[6] That work was in young children in diverse global settings, so the exact figures do not transfer to an adult in North Carolina, but the lesson does: **this test needs appropriate reference norms, and a single number without them is hard to interpret.**

The commercially available version, read honestly

A dual-sugar profile is commercially available as Genova Diagnostics' Intestinal Permeability Assessment, and its own sample report documentation supports an honest evaluation. It reports three values: lactulose percent recovery (reference at or below 1.50 percent), mannitol percent recovery (reference 4 to 27 percent), and the lactulose/mannitol ratio (reference at or below 0.10), calculated from pre-drink and post-drink urine levels, which corrects each result against your own baseline. The interpretation frame matches the physiology described above: elevated lactulose points at paracellular leak, elevated mannitol at transcellular permeability, low mannitol at malabsorption, and the ratio carries the headline signal, since it cancels the transit, hydration, and kidney factors that affect both sugars together. The report's own sample case shows why the ratio matters: both individual sugars within range while the ratio runs elevated.

Three transparency points, all from the vendor's own documentation, cited by name. The stated methodology is enzymatic rather than mass spectrometry, which is workable but is the softer quantification method, and it argues for reading single results conservatively. The reference ranges are population-statistical intervals (95 percent of a reference population), with Genova's own printed caution that values between one and two standard deviations are not necessarily abnormal and need clinical correlation. And like essentially every test in this category, it is a lab-developed test that has not been FDA-cleared, which the report states plainly.

The strongest use of this test in our model is baseline and retest, not one-time diagnosis. A single result inherits every caveat above. A change, measured by the same lab with the same assay and the same reference frame, before and after a defined eight-to-twelve-week intervention window, is the most defensible objective permeability evidence currently available to an outpatient, and it is the one thing no antibody panel or zonulin number can honestly provide. If you run it, run it twice, same lab both times, and let the delta, not the single number, carry the conclusion.

Multi-marker panels, where the evidence is better than for single markers

Combining markers does carry real information in defined clinical contexts. Serum biomarkers of intestinal barrier dysfunction have been studied for predicting postoperative endoscopic recurrence in Crohn's disease.[7]

That is the strongest argument for a panel, and also the clearest statement of its limits: the validation exists inside specific diseases with specific questions, not as a general leaky gut score.

What actually damages the barrier, which is more useful than measuring it

A review of human intestinal barrier function is unusually direct about this. **Stressors generally reduce barrier function**, and the named ones include nonsteroidal anti-inflammatory drugs, exercise, and pregnancy. Dietary factors that **increase** permeability include ethanol, fructose, and dietary emulsifiers. Factors reported to improve the barrier include vitamins A and D, tryptophan, cysteine, and fiber.[8]

Why I am putting this in the lab guide. For most people the actionable finding is not a number. It is that they take ibuprofen several times a week, drink most nights, and eat a diet built on emulsified processed food. Those are established barrier stressors and none of them requires a test to identify.

What to rule out before you conclude leaky gut

Permeability is a mechanism, not a diagnosis. These are diagnoses, they are treatable, and they get missed when everything is attributed to the barrier.

Worth ruling out	Why
Celiac disease	Real diagnosis, real treatment. **Test while still eating gluten**
Inflammatory bowel disease	Fecal calprotectin is the usual screen; blood in stool is never leaky gut
Small intestinal bacterial overgrowth	Bloating, distension, and irregularity overlap almost completely
Colorectal cancer	Which is why weight loss, bleeding, and anemia are red flags, not symptoms to interpret
Pancreatic insufficiency	Fatty stool, weight loss, and it is directly testable
Microscopic colitis	Watery diarrhea with a normal-looking colonoscopy unless biopsied
Bile acid diarrhea	Common, treatable, and routinely mistaken for IBS
Thyroid disease	Motility changes both directions

What to ask your clinician

- "Have I been screened for celiac disease, and do I need to be eating gluten for it to be valid?"
- "Would a fecal calprotectin be reasonable to rule out inflammatory bowel disease?"
- "I take NSAIDs regularly. Could that be contributing, and is there an alternative?"
- "If this permeability panel comes back positive, what would you change about my care?" The best question on the list, because if the answer is nothing, the test is not worth running.

Bring: a symptom timeline, your full medication list including over-the-counter NSAIDs, honest alcohol intake, and any previous results including ones you paid for yourself.

Reading your results as a pattern

Celiac serology positive. Stop here and pursue that. It is a real diagnosis with a treatment that works, and it outranks everything else on the page.

Calprotectin elevated. Needs gastroenterology. Not a supplement problem.

Barrier markers elevated, everything else negative, symptoms real. Reasonable to work on the modifiable stressors above. Not a reason for an open-ended protocol.

Everything negative, symptoms real. Common. It means the barrier hypothesis has not been supported, not that you are imagining anything, and the search should continue.

When to seek clinical review

Urgently: blood in stool, black tarry stool, unintended weight loss, persistent vomiting, fever with abdominal pain, or severe abdominal pain.

Promptly: new symptoms after starting a medication, or a change in bowel habit lasting more than a few weeks, particularly over age 45.

Before eliminating gluten: get celiac testing first, while still eating it.

Track what changes

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Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

The question list above also works in any office.

Get in touch: rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=gut-lab

Stress and sleep, which are barrier inputs rather than lifestyle advice

The reason these get their own section is that in the gut, unlike in most conditions, the stress effect has been measured directly rather than inferred.

Stress changes permeability, measurably, in healthy people. In an experimental study, acute stress produced a marked increase in intestinal macromolecular permeability, measured as albumin permeability, in healthy women but not in men. The authors had previously shown that chronic psychosocial stress induces jejunal epithelial barrier dysfunction, and they propose this mechanism as a contributor to female oversusceptibility to irritable bowel syndrome (PMID 22625665).

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- Five to ten minutes daily of slow breathing, longer exhale than inhale. Consistency beats technique.
- A fixed sleep window, seven to nine hours, same times.
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None of that costs anything, none of it interacts with anything, and leaving it out is why some people run three gut protocols and stay where they started.

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Seek care now for blood in the stool, black tarry stool, unintended weight loss, persistent vomiting, or fever with abdominal pain.

Get celiac testing before eliminating gluten, while you are still eating it. This comes up in every guide in this kit because it is the single most common self-inflicted mistake in gut health, and it costs you a clean answer permanently.

Remove before you add

Unlock Your Gut Key 2 lists what breaks the barrier in the first place: medications, dysbiosis, diet, stress, sleep, and infection. The lab guide turns that list into a table you can check yourself against. Work the ones that are yours, because a repair protocol that leaves the damage running is one you will repeat.

This is the organizing idea of the guide, and it is the opposite of how gut protocols are usually sold.

A review of human intestinal barrier function names the dietary factors that **increase** permeability: **ethanol, fructose, and dietary emulsifiers**. It names non-steroidal anti-inflammatory drugs among the stressors that reduce barrier function. And it lists factors that improve the barrier: vitamins A and D, tryptophan, cysteine, and fiber.[1]

Notice the asymmetry. The things that damage the barrier are specific, common, and mostly under your direct control. The things that improve it are ordinary nutrition. There is no exotic protocol in that list, on either side.

So the sequence is: stop the four documented stressors, feed the barrier ordinary food, and only then consider whether anything else is needed. That order is free, it is evidence-aligned, and it is the reverse of buying a repair supplement while the cause continues.

The four stressors, in order of how much control you have

Alcohol

Ethanol is named directly as increasing permeability.[1] This is not a moralizing point and I am not going to give you a safe number, because I do not have one for a person with gut symptoms.

What to do: a genuine four-week break, not a reduction, is one of the highest-yield experiments in this guide. If symptoms improve, you have learned something no panel would have told you.

NSAIDs

Ibuprofen, naproxen, and the rest are named among the stressors that reduce barrier function.[1] A great many people with gut symptoms take them several times a week for headaches, periods, or joint pain and never mention it, because it does not feel like medication.

What to do: count your actual weekly use honestly. If it is regular, that is a conversation with your prescriber about alternatives, not something to stop abruptly if you take them for a diagnosed condition.

Dietary emulsifiers

These are the additives that keep processed food smooth and stable: carboxymethyl cellulose, polysorbate 80, carrageenan, lecithins.

There is now human trial data. A placebo-controlled randomised trial put 60 healthy participants on an emulsifier-free diet for two weeks, then randomised them to four weeks of carboxymethyl cellulose, polysorbate 80, carrageenan, soy lecithin, native rice starch, or no additive, delivered in brownies.[2] After the two-week emulsifier-free run-in, cholesterol fell. Under supplementation, overall microbial diversity stayed stable but composition shifted by treatment, and short-chain fatty acid concentrations were lower with carrageenan.

Read that at the right size. It is 60 healthy people, four weeks, measuring mechanism rather than disease. It is not proof that emulsifiers cause illness. It is real human evidence that specific common additives change the gut environment measurably, in a design good enough to take seriously.

What to do: you do not need to memorise chemical names. Reducing ultra-processed food removes the great majority of dietary emulsifiers automatically, and does several other useful things at once.

Fructose

Named alongside ethanol and emulsifiers.[1] In practice this means sweetened drinks, high-fructose syrups, and large quantities of fruit juice rather than whole fruit.

What to do: cut liquid sugar first. It is the highest-dose, fastest-delivery form and the easiest to remove.

What a head-to-head found among the popular gut diets

Elimination diets are where most people spend their effort, so it is worth knowing how they compare.

Ten randomised trials covering 939 participants compared four dietary interventions for irritable bowel syndrome. The low FODMAP diet showed relatively consistent short-term efficacy, with responder rates of 34% to 78%. **In pooled analysis there was no significant difference versus the other dietary interventions**, with a risk ratio of 1.04, a confidence interval of 0.91 to 1.19, and no heterogeneity.[3]

Two things follow. Low FODMAP genuinely works for a substantial minority. And it is not better than the alternatives, which matters because it is by far the most restrictive and the hardest to sustain.

And it does not transfer to inflammatory bowel disease. A review of five randomised trials found **no effect of a low FODMAP diet on disease activity** in Crohn's disease or ulcerative colitis.[4] It may help IBS-type symptoms in someone who also has IBD; it does not treat the IBD.

How I would use that. If you try low FODMAP, run it as a structured short-term trial with a planned reintroduction, not as a permanent way of eating. It restricts many of the fermentable fibers that feed the microbiome, and the trial data does not support paying that cost indefinitely.

The foundation, which is ordinary and works

Fiber, named among the factors that improve barrier function.[1] Vegetables, legumes, whole grains if tolerated, fruit, nuts, and seeds. Increase gradually, because a fast jump causes the bloating people then blame on the food itself.

Protein sufficient for repair, which also supplies the tryptophan and cysteine named in the same review.[1] Eggs, fish, poultry, meat, legumes, dairy if tolerated.

Vitamin A from liver, eggs, dairy, and the orange and dark green vegetables. **Vitamin D** is difficult to obtain from food in useful amounts and is worth testing.

Fermented foods if you tolerate them. Yogurt, kefir, sauerkraut, kimchi. Reasonable, cheap, and pleasant, and their evidence is best treated as promising rather than settled.

Minimal ultra-processed food, which is how you remove emulsifiers, most added fructose, and a great deal else without reading a single label.

Running a fair experiment

Rule out first. Celiac and inflammatory bowel disease, per the lab guide. An elimination diet run over an undiagnosed celiac wastes months and destroys the test.

Then remove, one at a time.

1. **Liquid sugar and fructose**, two weeks. Fastest to show anything.
2. **Alcohol**, four weeks off, not reduced.
3. **NSAIDs**, if your use is regular, in consultation with whoever prescribes for you.
4. **Ultra-processed food**, four weeks, which handles emulsifiers.

Only then consider an elimination diet, and only as a structured trial with a reintroduction phase built in from the start.

Write down your baseline first. Gut symptoms fluctuate more than most, and without a record you will be comparing today against an inaccurate memory.

Track it: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=gut-diet

Realistic expectations

The four stressors are documented in human work, and removing them is free. That is the strongest thing this guide offers.

Low FODMAP helps 34% to 78% of people with IBS short term, and is no better than the alternatives.[3] It does not treat IBD.[4]

The foundation pattern is ordinary nutrition with reasonable support for barrier function.[1] It is not a repair protocol and no diet has been shown to seal an intestinal barrier on a timeline.

What I would not do is take on three simultaneous restrictions on the strength of a permeability panel whose limitations are covered in the lab guide.

When to seek clinical review

Urgently: blood in stool, black tarry stool, unintended weight loss, persistent vomiting, fever with abdominal pain.

Promptly: a change in bowel habit lasting more than a few weeks, particularly over age 45.

Before eliminating gluten: celiac testing, while still eating gluten.

Before a restrictive diet: if you are underweight, have a history of disordered eating, are pregnant, or are managing diabetes. Stacked eliminations carry real nutritional risk.

Working with Root Health

Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=gut-diet

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How you eat, and the two inputs that are not food

Digestion runs on the parasympathetic nervous system, so a meal eaten in a rush is processed differently from the same meal eaten sitting down. This is not a mood point, it is a mechanical one, and it costs nothing.

- Five slow breaths before the first bite, longer out than in.
- Sit down, no screens, no driving.
- Chew until the texture is gone.
- Stop at satisfied rather than full.

And the two inputs that no diet fixes. Acute stress produced a measurable increase in intestinal macromolecular permeability in healthy women in an experimental study, an effect not seen in men in the same experiment, building on earlier work showing chronic psychosocial stress induces jejunal barrier dysfunction (PMID 22625665). Sleep sits on the same gut-brain axis, with a 2026 review describing the bidirectional relationship while noting the mechanistic work is not finished (PMID 42092445).

If you change the food and nothing else, and stress and sleep are your actual barrier load, you will do this again next year.

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Seek care now for blood in the stool, black tarry stool, unintended weight loss, persistent vomiting, or fever with abdominal pain. Those are never a supplement problem.

Before anything in this guide: the lab guide in this kit lists the diagnoses that get missed when symptoms are attributed to the barrier. Celiac disease and inflammatory bowel disease both have real treatments and both are worth excluding before you spend money here.

Why nutrient evidence looks thinner than it is

A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. You cannot patent magnesium, an herb, or a food. So the evidence hierarchy this guide grades against is not neutral: it systematically over-documents patentable drugs and under-documents nearly everything else. That is measurable, not rhetorical. A Cochrane methodology review found that manufacturer sponsorship of drug and device studies produces more favourable results and conclusions than independent funding, through a bias that standard quality checks do not catch (PMID 28207928).

Two honest consequences follow, and they pull in opposite directions.

First, “no evidence” has three different meanings, and this guide tries to always tell you which one applies. *Tested and failed*: a real trial ran and the answer was no. *Never properly tested*: nobody could afford the trial, so absence of evidence is not evidence of absence. *Tested small and won*: real but modest trials, usually measuring markers rather than outcomes, because markers are what a small budget can afford.

Second, the asymmetry earns nutrients humility, not automatic credit. When isolated nutrients have received large publicly funded trials, the results have been mixed: some genuine wins, some nulls, and some harms. Small nutrient trials also carry their own publication bias toward positive results, just as drug trials tilt toward their sponsor. So this guide applies one rule in both directions: name the funding, name the evidence tier, and let you see both.

Stress and sleep, which are barrier inputs rather than lifestyle advice

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What to actually do, in Phase 1 alongside drainage rather than after it:

- Five to ten minutes daily of slow breathing, longer exhale than inhale. Consistency beats technique.
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Start with the finding that should reorder your spending

Glutamine is the flagship gut-repair supplement, and a meta-analysis found no overall effect on intestinal permeability.

A systematic review and meta-analysis of clinical trials on glutamine and gut permeability in adults found that **glutamine supplementation did not significantly affect intestinal permeability**, with a weighted mean difference of 0.00 and a confidence interval of 0.04 to 0.03.[1]

The abstract does report a positive subgroup at higher doses. I am not building a recommendation on it, because that abstract states the threshold as over 30 g per day in one sentence and over 30 mg per day in the next, and reports a confidence interval that does not contain its own point estimate. When a result is reported that inconsistently, the honest move is to note it and not lean on it.

And in inflammatory bowel disease specifically, a systematic review found glutamine supplementation had **no effect on disease course, intestinal permeability, intestinal morphology, disease activity, symptoms, biochemical parameters, oxidative stress, or inflammation markers.**[2] That is a comprehensive list of things it did not change.

The counterpoint, stated fairly. A review of human barrier function describes glutamine and zinc as among the substances that enhance the barrier and can reverse the effect of stressors.[3] That is a narrative review drawing on mechanistic and animal work alongside human data, and it sits in tension with the clinical-trial meta-analysis. Where a narrative review and a meta-analysis of trials disagree, I weight the trials.

What I would take from this. Glutamine is not dangerous and it is not the foundation of gut repair it is marketed as. If you are spending meaningfully on it, that money is better placed on the modifiable stressors in the diet guide.

Tier 1: The one category with real condition-specific evidence

Probiotics, with the emphasis on specificity

An umbrella review of probiotics, prebiotics, and synbiotics concluded that these agents offer **promising but selective clinical benefits depending on the condition and population**, are generally safe, and that their application **should be condition specific**.[4]

That last phrase is the entire recommendation. "Take a probiotic for gut health" is not a plan. "Take this strain for this condition" is, and only some conditions have that evidence.

Where the evidence is reasonable: irritable bowel syndrome has the largest body of work. In pediatric functional abdominal pain disorders, a systematic review found probiotics give **modest symptom relief** and should be considered a safe adjunctive therapy rather than a stand-alone treatment, in line with 2025 ESPGHAN and NASPGHAN guidelines, while noting that larger standardized trials are still needed to define strain, dose, and duration.[5]

Read "modest" as written. It is a real effect and it is not transformative.

Prebiotics, for contrast. The barrier review describes effects of prebiotics on barrier function as **modest**, while noting more documented effects for probiotics, especially combination agents, across in vitro, animal, and human work.[3]

Practically: pick a product with named strains at stated counts, matched to your actual condition rather than to the word "gut," give it four to eight weeks, and stop if nothing changes. A probiotic that has done nothing for three months is a subscription.

Zinc, if you are low

Named in the barrier review alongside glutamine as a substance that enhances barrier function and can reverse stressor effects.[3] Unlike glutamine, zinc deficiency is a real and testable state with consequences beyond the gut.

Functional target I use at Root Health: 70 to 110 ug/dL plasma.

Cautions: long-term zinc drives copper deficiency, which causes its own problems including a neuropathy that mimics B12 deficiency. Do not take it open-endedly without a reason. Separate from antibiotics and from thyroid medication.

Vitamins A and D, tryptophan, cysteine, and fiber

The same review lists these among dietary factors that improve the barrier.[3] Four of the five are food, and the diet guide in this kit covers them there, which is where they belong. Vitamin D is worth testing rather than guessing; the functional target is 40 to 80 ng/mL.

Tier 2: Ask your practitioner first

No products and no purchase links here.

Anything at all, if you have inflammatory bowel disease. IBD is a treated disease with real medications, and the glutamine result above is a specific warning against assuming supplements substitute. *Ask:* “Does anything I am taking interact with my IBD medication, and is there anything you would want me to avoid?”

High-dose zinc, long term. Because of copper. *Ask:* “Should we check copper if I stay on this?”

Anything, if you are on immunosuppressive therapy. Live probiotic organisms in an immunosuppressed person are a genuine clinical question rather than a formality.

If your symptoms have changed recently, or you are over 45 with a new change in bowel habit. That is a workup, not a supplement conversation.

Curcumin, with the trial that is usually oversold

The ebook names bioavailable curcumin in the barrier repair phase, so the honest evidence belongs here rather than in a footnote.

What the research shows. A randomised, double blind controlled trial assigned 206 patients with functional dyspepsia to curcumin, omeprazole, or both, for 28 days with follow-up to 56 days. All three groups improved significantly on dyspepsia scores, with no significant difference between them, no synergistic effect, and no serious adverse events (PMID 37696679).

The caveat that decides how much weight to give it. There was **no placebo arm.** Three active treatments, everyone improved, nobody differed. Functional dyspepsia has a large placebo response and a fluctuating natural course, so that design cannot separate “both treatments work” from “both are tracking placebo response and regression to the mean.” Also: 206 enrolled and 151 completed, and the paper carries a published correction from 2025 (PMID 38866470), which is a correction rather than a retraction.

And the population caveat, which matters here. That trial was in functional dyspepsia, not in intestinal permeability. There is no trial showing curcumin repairs a human gut barrier. Its inclusion rests on the anti-inflammatory mechanism described in the ebook plus this adjacent-condition trial, not on barrier outcome data.

Typical studied range. 250 mg four times daily, 2 g per day, in the trial above.

Formulation is the part people get wrong. Standard curcumin is poorly absorbed, which is why trials use enhanced delivery formats such as a phytosome or a piperine-paired preparation. Buying plain turmeric powder and expecting trial results is the common mistake.

Cautions. Increases bleeding risk with anticoagulants and antiplatelet drugs, stop before surgery. Can cause reflux and stomach upset. Interacts with several drugs through the same liver enzymes. Not established in pregnancy.

What I deliberately left out, and why

Most of the gut repair category. Collagen, bone broth, aloe, slippery elm, deglycyrrhizinated licorice, and the various proprietary barrier blends. Some may help. I did not find human trial evidence that they change permeability or outcomes, and I am not going to imply otherwise.

Anything sold as sealing or healing the gut lining on a stated timeline. No supplement has demonstrated that.

Proprietary blends with undisclosed amounts. You cannot compare them to a studied dose or tell a clinician what you are taking.

Things I use in practice that are not in here. A guide sold to someone I have never examined has no history, no medication list, and no way to course-correct. The bar for a document is higher than the bar for a conversation. Where the evidence does not support recommending something at a distance, it stays out, and that is exactly why you can trust what did make it in.

Running a fair trial

Rule out first. Celiac and IBD, per the lab guide. Everything else is guessing until those are excluded.

Remove before you add. The diet guide covers the stressors with actual human evidence: alcohol, NSAIDs, fructose, and emulsifiers. Removing a documented cause beats adding a speculative fix, and it is free.

One at a time, four to eight weeks, with a written baseline.

Track it: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=gut-supplement

Where to find quality versions of these

Everything above was written without reference to any product line.

Third-party testing first. NSF Certified for Sport and USP Verified mean an independent lab confirmed the contents. **For probiotics specifically, look for named strains with counts guaranteed through the expiry date**, not at manufacture, since that is where cheap products fail.

us.fullscript.com/welcome/drseay

Disclosure: Root Health earns a portion of sales through that link. Nothing here was included, excluded, ranked, or dosed based on what it earns. The best-selling supplement in this entire category, glutamine, is the one this guide opens by telling you did not beat placebo.

When to seek clinical review

Urgently: blood in stool, black tarry stool, unintended weight loss, persistent vomiting, fever with abdominal pain.

Promptly: a change in bowel habit lasting more than a few weeks, particularly over age 45.

Before starting: if you take immunosuppressive medication, or have IBD.

Working with Root Health

Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=gut-supplement

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Get in touch: rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=gut-supplement

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Functional target ranges transcribed from the Root Health blood chemistry range.

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